ExamNotes AI

Sub Topics (Priority Wise)

High Priority:
• Introduction to Organic Chemistry
• Functional Groups and Nomenclature
• Reactions of Carbonyl Compounds

Medium Priority:
• Reactions of Alkanes
• Reactions of Alkenes
• Reactions of Alcohols and Ethers

Detailed Notes

1. Introduction to Organic Chemistry (high)
Organic chemistry deals with carbon containing compounds. It is the core of life sciences and industry. Key concepts include the C-H bond, functional groups, and the importance of isomerism.

2. Functional Groups and Nomenclature (high)
Functional groups determine the reactivity of organic molecules. Major groups: alkanes, alkenes, alkynes, alcohols, ethers, aldehydes, ketones, carboxylic acids, esters, amides. IUPAC rules give systematic names.

3. Reactions of Alkanes (medium)
Alkanes undergo free radical halogenation (Cl2, Br2) and combustion. Halogenation is controlled by light and temperature; products are mixtures of alkyl halides.

4. Reactions of Alkenes (medium)
Alkenes undergo electrophilic addition (hydrogen halides, H2O, HBr, H2SO4). They also polymerize (radical, cationic, anionic). Regio- and stereochemistry (E/Z) are important.

5. Reactions of Carbonyl Compounds (high)
Aldehydes and ketones react in nucleophilic addition (Grignard, cyanide). Carboxylic acids form derivatives (esters, amides) and undergo nucleophilic acyl substitution.

6. Reactions of Alcohols and Ethers (medium)
Alcohols can be oxidized, dehydrated, or substituted. Ethers are cleaved by strong acids or via Williamson ether synthesis.

Revision Points

• Alkanes are saturated hydrocarbons.
• Alkenes have C=C double bonds.
• Aldehydes have R-CHO, ketones R-CO-R'.
• Carboxylic acids are R-COOH.

• Ethers have R-O-R'.
• SN1 involves carbocation intermediate.
• SN2 is a concerted backside attack.
• Addition reactions add across double bonds.
• Elimination reactions remove H and halogen.
• Oxidation of primary alcohol gives carboxylic acid.
• Reduction of carbonyl gives alcohol.
• Grignard reagents add to carbonyl.
• Esterification forms esters.
• Acid-catalyzed dehydration of alcohols gives alkenes.

Important Questions

Short Questions:
• Explain the difference between SN1 and SN2 mechanisms.
• What is the major product of the addition of HBr to propene?
• Describe the steps in the Williamson ether synthesis.
• What is the role of a catalyst in the polymerization of ethylene?
• Name two common reagents used for the oxidation of primary alcohols.

Long Questions:
• Discuss the mechanism of free radical halogenation of alkanes, including initiation, propagation, and termination steps.
• Explain the stereochemical outcome of the addition of H2SO4 to 2-butene and how it leads to Markovnikov product.
• Compare the reactivity of aldehydes and ketones towards Grignard reagents and justify the differences.

Diagram Question:
Draw the mechanism of a nucleophilic substitution (SN2) of a primary alkyl halide with a hydroxide ion.